

# Akhil Chigurupati

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## PROFESSIONAL SUMMARY

Software engineer with 1+ years of professional experience building AI-powered systems and full-stack applications, including LLM-powered RAG pipelines, autonomous AI agents, MCP tool servers, real-time Kafka data pipelines, and React/TypeScript frontends backed by FastAPI and AWS. Production experience managing 500+ services at TCS. Independently shipped four projects spanning agentic AI orchestration, distributed analytics, and ML research published in Springer. Comfortable across the full stack: React frontends, Python and Node.js backends, PostgreSQL, Docker, Kubernetes, and CI/CD pipelines.

## PROFESSIONAL EXPERIENCE

**Tata Consultancy Services** | Software Engineer | Apr 2023 – May 2024

- Developed Python microservices and REST APIs for an AI-driven infrastructure monitoring platform managing **500+ production services**, reducing debug cycle time through clean component architecture and well-defined service boundaries.
- Architected **Apache Kafka** streaming pipelines processing **50,000+ events/min** for real-time log and metrics ingestion, reducing alert latency from minutes to seconds and enabling pre-impact incident detection.
- Built and integrated **AI/ML anomaly detection models** into production monitoring workflows across 500+ services, flagging failure patterns **8-12 minutes** before user-facing degradation.
- Deployed containerized applications on **Docker and Kubernetes** across **1,000+ pods** with auto-scaling; automated incident remediation workflows reducing manual resolution effort by **30-40%**.

## PROJECT EXPERIENCE

**MCP Tool Server & Autonomous Agent | Agentic AI Platform with Tool Orchestration**

*Python • Anthropic MCP SDK • Model Context Protocol • Groq API • LLaMA 3 • FastAPI • PostgreSQL • Docker Compose • GitHub Actions*

- Designing and building a custom **MCP (Model Context Protocol) server** using the Anthropic MCP SDK, exposing multiple tools (database queries, ticket creation, external API integrations) callable by Claude Desktop and a custom autonomous agent client.
- Building an **autonomous agent client** powered by Groq API (LLaMA 3) that connects to the MCP server, orchestrates multi-step tool calls autonomously, and implements an **error recovery and retry loop** with all agent decisions and tool calls logged and persisted in PostgreSQL.

**AI Incident Intelligence Platform | LLM-Powered Incident Analysis & Response**

*Python • FastAPI • LangChain • LlamaIndex • OpenAI API • RAGAS • MLflow • Supabase pgvector • React • TypeScript • AWS (Lambda, SQS, DynamoDB, SNS) • Docker • GitHub Actions*

- Architected an event-driven incident intelligence platform ingesting real-time metrics via **AWS SQS**, detecting anomalies with LLM-powered analysis, and streaming remediation recommendations to a **TypeScript React** frontend via SSE. LangChain retrieval chains and LlamaIndex index historical incidents in Supabase pgvector to ground every response.
- Built a **RAGAS evaluation pipeline** automatically scoring every RAG response for faithfulness, answer relevancy, and context precision; integrated **MLflow** to log eval scores across iterations, enabling measurable, tracked improvement in response quality.

**URL Shortener | Distributed Analytics Platform**

*Python • FastAPI • PostgreSQL • Redis • Apache Kafka • Prometheus • Grafana • Docker Compose • GitHub Actions • pytest*

- Built a full-stack distributed analytics platform with async **Kafka consumer workers** decoupling click-event tracking from the redirect path; cache-aside pattern with **Redis** for sub-millisecond lookups; instrumented with Prometheus/Grafana; full CI/CD via GitHub Actions with pytest.

**Review Reality | Fake Review Detection - Springer Published (Feb 2025)**

*Python • Scikit-learn • Random Forest • Naive Bayes • MLP • Voting Classifier • NLP • Sentiment Analysis • Amazon/Yelp Dataset*

- Developed ensemble ML algorithms (Random Forest, Naive Bayes, MLP, Voting Classifier) achieving **92% precision** across **500,000+ entries** on 10 platforms with advanced NLP feature engineering; reduced false positive rate by 30% and improved fraud identification by 15%, validated in **peer-reviewed Springer publication**.

## EDUCATION

**Regis University** - Master of Science, Information Systems | Denver, CO

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL, Java, C, HTML, CSS

**Frontend:** React.js, TypeScript, Vite, Server-Sent Events (SSE), HTML, CSS

**Backend:** FastAPI, Node.js, REST APIs, Microservices, API Development, System Design, Apache Kafka, AWS SQS

**AI/ML & Agentic:** LangChain, LlamaIndex, MCP SDK, Model Context Protocol, RAG, RAGAS, MLflow, Autonomous Agent Loops, LLM Tool Calling, ReAct Agent Pattern, Multi-Step Task Execution, OpenAI API, Claude API, Groq API, LLaMA 3, sentence-transformers, Scikit-learn, TensorFlow, Anomaly Detection

**Cloud & DevOps:** AWS (Lambda, SQS, DynamoDB, API Gateway, SNS, CloudWatch), Docker, Kubernetes, Helm, GitHub Actions, GitLab CI/CD, Linux, CI/CD Pipelines

**Databases:** PostgreSQL, Redis, MongoDB, Supabase pgvector

**Monitoring:** Prometheus, Grafana, ELK Stack, MLflow, Infrastructure Monitoring

**Core Competencies:** Distributed Systems, Agile Development, Data Structures, Algorithms, Root Cause Analysis, Debugging, Incident Response, System Design